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129 Citations

# Tzu-Han Hsu



## RESEARCH INTERESTS

My research focuses on developing algorithms for **verification** and **synthesis** using **formal methods** within the relational reasoning framework named **hyperproperties**, including applications in **privacy-aware** policies, **secure information-flow**, **reinforcement learning**, and **safety-guaranteed** analysis. My current work explores **integrating AI and large language models (LLMs) with formal methods**, bringing together the reasoning power of modern AI and the rigor of formal guarantees.

## EDUCATION

**Ph.D. in Computer Science and Engineering** 2020 – 2026

Michigan State University, East Lansing, MI

**GPA:** 3.93/4.0      **Advisor:** Dr. Borzoo Bonakdarpour

🏆 2026 Three Minute Thesis (3MT) Competition Winner, MSU College of Engineering

🏆 2024 Best Research Poster Award, MSU College of Engineering

**M.S. in Computer Science and Engineering**

Michigan State University, East Lansing, MI

**GPA:** 3.93/4.0

🏆 2021 Google Research Fellowship Nomination, MSU College of Engineering

**B.S. in Computer Science** 2016 – 2020

Iowa State University, Ames, IA

**GPA:** 3.86/4.0

🏆 Graduated *Magna Cum Laude*

🏆 2013 – 2020 ISU College of Liberal Arts and Sciences Dean's List

**B.M. in Piano Performance** 2013 – 2017

Iowa State University, Ames, IA

**GPA:** 3.86/4.0

🏆 Outstanding Senior Nomination, ISU Department of Music

## RESEARCH EXPERIENCE

**Postdoctoral Research Fellow @ Harvard University** 2026 – Present

**Advisor** Prof. Nada Amin

**Topic** INTEGRATING AI AND LARGE LANGUAGE MODELS (LLMs) WITH FORMAL METHODS

## Graduate Research Assistant @ Michigan State University

2020 – 2026

**Advisor** Prof. Borzoo Bonakdarpour

**Topic** REASONING ABOUT HYPERPROPERTIES THROUGH SYMBLIC DECISION PROCEDURES

Keywords: HyperLTL, SMT Solving, Symbolic Execution, SyGuS, Runtime Enforcement, Program Repair, Policy Synthesis, Model Checking, Reinforcement Learning

## Research Intern @ Microsoft Research (MSR)

Summer 2023

**Mentors** Shuo Chen, Nikolaj Bjørner

**Project** THEOREM-CARRYING-TRANSACTION: PROVABLE SECURITY FOR SMART CONTRACTS

Keywords: Solidity, Boogie, EVM, Go-Ethereum, ERC20, UniSwap-V2

## Applied Scientist Intern @ Amazon AWS Automated Reasoning Group

Summer 2022

**Mentors** Aman Goel, Michael Whalen

**Project** SYMBOLIC MODEL CHECKING FOR DISTRIBUTED PROTOCOLS WITH PARALLELIZATION

Keywords: P Language, Symbolic Execution, AWS Lambda, Amazon S3

## Undergraduate Research Assistant @ Iowa State University

2019 – 2020

**Advisor** Prof. Borzoo Bonakdarpour

**Project** BOUNDED MODEL CHECKING FOR HYPERPROPERTIES

Keywords: QBF/SMT Solvers, Z3, QuAbS, Linear Temporal Logic, Information-Flow Security

## JOURNAL PAPERS

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- [j1] Tzu-Han Hsu, Ana Oliveira da Costa, Andrew Wintenberg, Ezio Bartocci, Borzoo Bonakdarpour  
“**Gray-box Runtime Enforcement of Hyperproperties**”  
*Acta Informatica* (2025).

## CONFERENCE PAPERS

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- [c1] Tzu-Han Hsu, Milad Rabizadeh, Kenneth Rogale, Fedor Filippov, Marco A. de Oliveira Batista, and Borzoo Bonakdarpour  
“**HyperQB2.0: A Bounded Model Checker for Hyperproperties**”  
*International Conference on Computer Aided Verification (CAV’26)*, to appear.
- [c2] Tzu-Han Hsu, Arshia Rafeioskouei, Borzoo Bonakdarpour  
“**HypRL: Reinforcement Learning of Control Policies for Hyperproperties**”  
*Annual Conference on Neural Information Processing Systems (NeurIPS’25)*, San Diego, CA.
- [c3] Raven Beutner, Tzu-Han Hsu, Borzoo Bonakdarpour, and Bernd Finkbeiner  
“**Syntax-Guided Automated Program Repair For Hyperproperties**”  
*International Conference on Computer Aided Verification (CAV’24)*, Montreal, Canada.
- [c4] Tzu-Han Hsu, Borzoo Bonakdarpour, Bernd Finkbeiner, and César Sánchez  
“**Bounded Model Checking for Asynchronous Hyperproperties**”  
*International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS’23)*, Paris, France.
- [c5] Tzu-Han Hsu, César Sánchez, Sarai Sheinvald, and Borzoo Bonakdarpour  
“**Efficient Loop Conditions for Bounded Model Checking Hyperproperties**”  
*International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS’23)*, Paris, France.

- [c6] [Tzu-Han Hsu](#), Borzoo Bonakdarpour, Eunsuk Kang, and Stavros Tripakis  
**“Mapping Synthesis for Hyperproperties”**  
*IEEE International Symposium on Computer Security Foundations (CSF’22/FLoC’22)*, Haifa, Israel.
- [c7] [Tzu-Han Hsu](#), César Sánchez, and Borzoo Bonakdarpour  
**“Bounded Model Checking for Hyperproperties”**  
*International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS’21)*, Luxembourg, Luxembourg.

## ARXIV PREPRINTS

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- [a1] Thomas Ball, Nikolaj S. Bjørner, Ashley J. Chen, Shuo Chen, Yang Chen, Zhongxin Guo, [Tzu-Han Hsu](#), Peng Liu, Nanqing Luo  
**“Theorem-Carrying Transactions: Runtime Verification to Ensure Interface Specifications for Smart Contract Safety (v2)”**
- [a2] [Tzu-Han Hsu](#), Yu Wang, Borzoo Bonakdarpour, Miroslav Pajic  
**“Multi-Agent Path Planning with Hyperproperties (v1)”**

## RESEARCH FELLOWSHIPS

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- **Michigan State University Graduate Research Fellowship** 2021, 2022  
 Received Summer Support for outstanding students selected by MSU CSE department.
- **Google Research Fellowship Nomination** 2021  
 Selected as one of the only four nominees selected by MSU College of Engineering.

## RESEARCH AWARDS

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- **Three Minute Thesis (3MT) Competition Winner** 2026  
 Awarded by the Michigan State University College of Engineering.
- **Best Research Poster Award** 2024  
 Awarded by Michigan State University College of Engineering Graduate Research Symposium.

## INVITED RESEARCH TALKS

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- **Purdue University Programming Languages Group (PurPL) Seminar** 2025  
*Hosted by prof. Benjamin J Delaware*
- **Academia Sinica (National Academy of Scientific Research, Taiwan)** 2023  
*Hosted by prof. Yu-Fang Chen*
- **Portland State University Department of Computer Science PLV Seminar** 2022  
*Hosted by prof. Yao Li*
- **University of California Santa Cruz: Languages, Systems, and Data Seminar** 2021  
*Hosted by prof. Lindsey Kuper and prof. Tyler Sorensen*
- **RWTH Aachen University, Germany** 2021  
*Hosted by prof. Joost-pieter katoen and prof. Erika Ábrahám*

## PROFESSIONAL SERVICES

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### · Program Committee:

|               |                                                                          |     |
|---------------|--------------------------------------------------------------------------|-----|
| <b>OOPSLA</b> | <i>ACM SIGPLAN Conf. on OOP, Systems, Languages &amp; Applications</i>   | '27 |
| <b>EMSOFT</b> | <i>International Conference on Embedded Software</i>                     | '26 |
| <b>LMPL</b>   | <i>ACM SIGPLAN Workshop on Language Models and Programming Languages</i> | '26 |
| <b>EMSOFT</b> | <i>International Conference on Embedded Software, WiP-LB TPC</i>         | '25 |

### · Conference Paper Reviews:

|               |                                                                              |         |
|---------------|------------------------------------------------------------------------------|---------|
| <b>CSL</b>    | <i>European Association for Computer Science Logic</i>                       | '25     |
| <b>SRDS</b>   | <i>International Symposium on Reliable Distributed Systems</i>               | '24     |
| <b>EMSOFT</b> | <i>International Conference on Embedded Software</i>                         | '24     |
| <b>CAV</b>    | <i>International Conference on Computer Aided Verification</i>               | '24     |
| <b>RV</b>     | <i>International Conference on Runtime Verification</i>                      | '24,'25 |
| <b>TACAS</b>  | <i>Tools and Algorithms for the Construction and Analysis of Systems</i>     | '23,'25 |
| <b>FM</b>     | <i>International Symposium on Formal Methods</i>                             | '23     |
| <b>ICCPS</b>  | <i>International Conference on Cyber-Physical Systems</i>                    | '23     |
| <b>VSTTE</b>  | <i>Verified Software: Theories, Tools, Experiments</i>                       | '23     |
| <b>NFM</b>    | <i>NASA Formal Methods</i>                                                   | '23     |
| <b>DISC</b>   | <i>International Symposium on Distributed Computing</i>                      | '22     |
| <b>CSF</b>    | <i>Computer Security Foundations Symposium</i>                               | '22     |
| <b>ATVA</b>   | <i>Int'l Symposium on Automated Technology for Verification and Analysis</i> | '21     |

### · Conference Artifact Evaluation:

|            |                                                                |         |
|------------|----------------------------------------------------------------|---------|
| <b>CAV</b> | <i>International Conference on Computer Aided Verification</i> | '23,'24 |
|------------|----------------------------------------------------------------|---------|

### · Journal Reviews:

|             |                         |             |
|-------------|-------------------------|-------------|
| <b>ACTA</b> | <i>Acta Informatica</i> | '22,'25,'26 |
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## TEACHING EXPERIENCE

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- **Graduate Teaching Assistant** Spring 2020  
Michigan State University, CSE 260: Discrete Structures in Computer Science
- **Undergrad Teaching Assistant** Spring, Fall 2018  
Iowa State University, COM S 227: Introduction to Object-Oriented Programming

## HONORS AND SCHOLARSHIPS

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- **Outstanding Senior Nomination** by ISU Music Department 2020
- **Dean's List** of Iowa State University College of Liberal Arts and Sciences 2013–20
- **Future Leader Award** issued by U.S. Ambassador to China Terry Branstad 2018
- **Gold Level Scholarship Recipient** as Iowa State University International Student Ambassador 2013–18
- **Outstanding Student Scholarship Recipient** by Iowa State University Music Department 2013–18

## LEADERSHIP AND VOLUNTEER EXPERIENCES

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- **Research Conference Student Volunteer** at Computer Aided Verification for information publicity 2021

- **Certificate of Achievement** from The Society for Collegiate Leadership & Achievement 2016–20
- **Department Student Representative** of Iowa State University Student Council 2017–18
- **Volunteer/Event Coordinator** at Iowa Education without Borders (non-profit organization) 2017–18
- **Student Mentor** at Iowa Youth Leadership Summer Camp 2017–18
- **Student Volunteer** at Iowa State University Annual K-12 Computational Thinking Competition 2020

## MUSIC AWARDS AND HONORS

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- **Assistant Concertmaster (Violin)** at Iowa State University Symphony Orchestra 2015-17
- **Soloist Winner - Piano Division** Fort Dodge Orchestra Young Artists Concerto Competition 2015
- **1st Prize Winner** Iowa State University Symphony Orchestra Concerto Competition 2014
- **State-Level Competition Winner** Music Teacher National Association (MTNA) Piano Competition 2014

## INDUSTRIAL EXPERIENCE

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**Software Engineer Intern** 2019–20

Worrell Medical Design Inc., Minneapolis, MN

- Developed a Virtual Reality Environment on *Oculus Quest* for usability testing in Human Factors Engineering.
- Programmed well-functioning C# scripts in *Unity3D* engine with desired behaviors and outcomes.
- Analyzed user feedback and debugged/adjusted/improved the existing versions of the program with human factors engineers.